

Security incident report

Section 1: Identify the network protocol involved in the incident

Based on the incident, DNS and HTTP are core network protocols that the malicious actor has attempted to alter.

Section 2: Document the incident

The website "yummyrecipesforme.com" has been hacked into this website through brute force method, where the former employee has attempted multiple ways to guess username, password, and other sensitive information. Based on tcpdump traffic log, when the source computer is requesting data from the dns server "dns.google.domain" which refers to website "yummyrecipesforme.com" that browser has successfully established a connection through ACK packet. But later, the IP address has been spoofed as the destination IP address has changed from "yummyrecipesforme.com" to "greatrecipeforme.com.http" which corresponds to IP address "192.0.2.172".

Therefore, the network traffic has changed between source computer and spoofed website "greatrecipeforme.com.http" and the associated port number is "56378"

Due to this tcpdump traffic log, many customers complained about the website, telling them to download a file to access the recipe. But under the hood, the website IP address has changed to "192.0.2.172". Because of that change in IP address, after downloading and running the file, the address of the website changed and their personal computers began running more slowly.

Section 3: Recommend one remediation for brute force attacks

In order to remediate for brute force attacks, you create a sandbox environment which sort of behaves like a VM/Virtual Machine. Another way to mitigate this issue is to initiate a tcp network protocol analyzer.

Once you are ready with the sandbox, you start going to "yummyrecipesforme.com" and as soon as the website loads, you are prompted to download an executable file to update your browser. You accept the download and allow the file to run. You then observe that your browser redirects you to a different URL, greatrecipesforme.com, and later reboot the whole system, or block the website to look further into IP address, or change either one of username, passwords, and reset it to a new information, so the malicious actor cannot guess at all.